

On general visibility graph recognition

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Basic definitions

Definition

A **polygon** is a cyclically ordered set of points in a plane, with no three successive points collinear, together with the line segments joining consecutive pairs of the points.

Let A and B be arbitrary points of a polygon. We say that the pair (A,B) is **visible** if the line segment AB is a side of the polygon, or if AB lies entirely within the polygon. figure here



Visibility graph construction

Given a polygon of n points, construct a graph with n vertices—each corresponding to a point. Draw an edge between two vertices if the corresponding pair is visible. The resulting graph is the **visibility graph**.
figure here

Pocchiola and Vegter give an algorithm with $O(n \log n + E)$ time complexity and $O(n)$ space complexity.



Problem description

A seminal expository paper by Ghosh and Goswami lists several related open problems in the field of computational geometry.

Open problem 28 (*visibility graph recognition*)

Given a graph G in adjacency matrix form, determine whether G is the visibility graph of a simple polygon P .

In essence, we are tasked to compute the reverse direction.



Open problems

Open problem 29

Is the problem of recognizing visibility graphs in NP?

Open problem 32

Characterize visibility graphs of simple polygons.

Open problem 33

Draw a simple polygon whose visibility graph is the given graph G .

We propose a construction which resolves all of the above open problems.



An iterative construction

Notation

$Vis(n)$ refers to the class of all non-isomorphic visibility graphs of size n .

For a given graph G of size s , we set out to generate all distinct visibility graphs of size less than or equal to s . The algorithm can be bounded in exponential time complexity.



An iterative construction

Sew operation

Now we describe the **sew operation**, which takes a polygon of n points and returns a polygon of $n + 1$ points. Given a point C outside the polygon, we "sew" it into the polygon by breaking an existing edge AB and adding the edges AC, BC .



An iterative construction

Our approach is inductive. Begin by considering the base case of $Vis(3)$, which consists of the single element K_3 . K_3 is associated with all non-degenerate triangles.

The inductive step is as follows. Given a visibility graph and its corresponding polygon, draw a line between all pairs of points in the polygon. This partitions the plane into regions.

Claim

Let A, B be distinct points lying in the same region. Perform the sew operation on points A and B to obtain polygons P_A and P_B . $Vis(P_A)$ and $Vis(P_B)$ are isomorphic.



A lemma on triangulations

Lemma

All possible triangulations of polygons contain at least one triangle consisting of 3 cyclically adjacent vertices.

The proof of this is trivial.



Chop operation

Chop operation

Now we describe the **chop operation**, which takes a polygon of n points and returns a polygon of $n - 1$ points. Given a point B of the polygon, we "chop" B out of the polygon by breaking the edges AB, BC and inserting the edge AC .

We claim that for all polygons, it is possible to select a point on the polygon on which the chop operation can be performed. This follows naturally from the stated lemma.



Meet semilattices

Definition

A poset $(P, \leq)$ is called a **meet semilattice** if every subset of two elements in P has a greatest lower bound.

Consider arbitrary $p, q \in P$. Let x be an element of P such that $x \leq p$ or $x \leq q$. If there are only finitely many such candidates for x , the existence of a minimal element $m \in P$ implies that P is a meet semilattice (dual of Zorn's lemma).



Meet semilattices and Vis

Theorem

The poset $(Vis, \leq_S)$ is a meet semilattice, where $\leq_S$ denotes the induced subgraph operator. Furthermore, there is a bijective correspondence between the maximal antichains of $(Vis, \leq_S)$ and the equivalence classes of Vis , sorted by graph size.



Known bounds on Vis

Theorem (Warren)

For m distinct n -degree polynomial equations of maximum degree d , the number of possible sign patterns is at most $(\frac{4edm}{n})^n$.

Spinrad describes a indirect application of Warren's theorem to upper bound $|Vis(n)|$ by $2^{O(n \log n)}$. A later paper by Ghosh proved that 2^{n-4} is a lower bound for $|Vis(n)|$.



Conjectures and more

We end by proposing a variety of open problems which illuminate current understanding of visibility structures.

Open problem

Find a more efficient algorithm to solve the same problem.

Open problem

Characterize the meet semilattice $(Vis, \leq_S)$. Is there any connection with known partial orders? What do the meet and join operations represent?

Open problem

Is there a generalization for visibility graphs of n -dimensional polytopes?



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